

Hands on 4 - Difference between JPA, Hibernate and Spring Data JPA

Employee.java:

```
package com.example.demo.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;

@Entity
public class Employee {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Integer id;

    private String name;

    private double salary;

    public Employee() {
    }

    public Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }

    public Integer getId() {
        return id;
    }

    public void setId(Integer id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public double getSalary() {
        return salary;
    }
}
```

```
    public void setSalary(double salary) {  
        this.salary = salary;  
    }  
}
```

application.properties:

```
spring.datasource.url=jdbc:h2:mem:testdb  
spring.datasource.driverClassName=org.h2.Driver  
spring.datasource.username=sa  
spring.datasource.password=
```

```
spring.jpa.hibernate.ddl-auto=update
```

```
spring.jpa.show-sql=true
```

```
spring.h2.console.enabled=true
```

EmployeeRepository.java:

```
package com.example.demo.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository;  
import com.example.demo.entity.Employee;
```

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```

EmployeeService.java:

```
package com.example.demo.service;
```

```
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.stereotype.Service;  
import org.springframework.transaction.annotation.Transactional;
```

```
import com.example.demo.entity.Employee;  
import com.example.demo.repository.EmployeeRepository;
```

```
@Service  
public class EmployeeService {
```

```

@Autowired
EmployeeRepository repository;

@Transactional
public void addEmployee(Employee employee) {

    repository.save(employee);

}

}

```

Output:

```

2026-07-09T13:19:32.914+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] c.c.s.DemoApplication : Starting DemoApplication using Java 26 with PID 17055 (/Users/adityahargunani/Downloads/spring-data-jpa-demo)
2026-07-09T13:19:32.915+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] c.c.s.DemoApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-09T13:19:33.098+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-07-09T13:19:33.114+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 14 ms. Found 1 JPA repository interface.
2026-07-09T13:19:33.242+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-09T13:19:33.247+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-09T13:19:33.247+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-09T13:19:33.264+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 333 ms
2026-07-09T13:19:33.321+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] org.hibernate.orm.jpa : HH000540: Processing PersistenceUnitInfo [name: default]
2026-07-09T13:19:33.327+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] org.hibernate.orm.core : HH000001: Hibernate ORM core version 7.4.1.Final
2026-07-09T13:19:33.462+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup; ignoring JPA class transformer
2026-07-09T13:19:33.477+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-09T13:19:33.536+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:testdb user=SA
2026-07-09T13:19:33.537+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-07-09T13:19:33.537+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] org.hibernate.orm.connections.pooling : HH010001005: Database info:
Database JDBC URL [jdbc:h2:mem:testdb]
Database driver: H2 JDBC Driver
Database dialect: H2Dialect
Database version: 2.4.240
Default catalog/schema: TESTDB/PUBLIC
Autocommit mode: undefined/unknown
Isolation level: READ_COMMITTED (default READ_COMMITTED)
JDBC fetch size: 100
Pools DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-07-09T13:19:33.583+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] org.hibernate.orm.core : HH0000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
Hibernate: create table employee (id integer generated by default as identity, name varchar(255), salary float(53) not null, primary key (id))
2026-07-09T13:19:33.624+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-07-09T13:19:33.856+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-07-09T13:19:33.911+05:30 WARN 17055 --- [spring-data-jpa-demo] [main] JpaBaseConfigurationJpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-07-09T13:19:34.087+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.s.b.h.a.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available at 'jdbc:h2:mem:testdb'
2026-07-09T13:19:34.025+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-09T13:19:34.026+05:30 INFO 17055 --- [spring-data-jpa-demo] [main] c.c.s.DemoApplication : Started DemoApplication in 1.239 seconds (process running for 1.378)
Employee Saved

```

